



THE UNIVERSITY OF
SYDNEY

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SCHOOL OF AEROSPACE, MECHANICAL AND MECHATRONIC
ENGINEERING

AERO2460: AEROSPACE DESIGN 1

ASSESSMENT 1B: GROUP DESIGN ASSIGNMENT (PART 2)

Design of an L-Bracket for Aerospace and Automotive Applications

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August 4, 2026

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Abstract

The purpose of this project is to design, develop, and test two 3D-printed L-brackets for both an aerospace and automotive application. These designs were then tested in a UTM and compared to a benchmark design. Both designs outperformed the benchmark.

The aerospace bracket was shaped to carry load mainly through a triangulated, axial path to improve stiffness and capacity. The automotive bracket kept a simpler brace layout to shorten print time while still lifting performance over the benchmark. All brackets were printed to a common baseline and tested under displacement control using a standard cantilever setup. We recorded peak load and tip deflection, and used a cost metric to weigh performance against print time for each application.

Results show both concepts exceeded the benchmark in strength and stiffness. Failures started at the brace/strut roots, which is consistent with stress concentrations. Overall, the aerospace bracket is stronger but takes longer to print, while the automotive bracket is a good choice when faster printing is the priority.

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In Figure ??, the load-displacement curve peaks and then decreases slowly, indicating progressive damage rather than sudden failure. The in-jig and post-test failure graphics (Figure ??) confirm that the first damage initiated at the bases of the V-struts where they form a junction with the horizontal flange. Under the applied end load, the V-struts were in axial tension on the underside of the V-struts/flange and pulled upward at their roots. As this tensile force was transferred into the flange over a short filleted run-out, the junction experienced a combination of net-section tension and peel (opening) plus in-plane shear, typical of a geometry change where an axial member anchors into a plate. In the preliminary design phase, the addition of the 2 mm fillets sought to mitigate the stress concentration. However, ?? indicates that it did not eliminate this entirely.

The fracture surfaces show layer-by-layer opening with limited ductility, consistent with FDM PLA's anisotropy and interlayer adhesion limits. At the strut-flange root, the tensile strut load is turned into the flange, creating mixed-mode loading: Mode I opening/peel (layers pulled apart) together with Mode II in-plane shear (layers sliding). Under this combination, cracks preferentially follow print interfaces [5, 6]. This explains the rounded post-peak softening in Figure ??: as microcracks formed at one strut root, a residual ligament and the opposite strut continued to carry load, giving a gradual loss of load-carrying capacity rather than instantaneous failure. Mechanically, the V-struts achieved their purpose of increasing stiffness and capacity, but the governing section remained the strut-flange root junction in tension.

From a safety perspective, the behaviour shows some damage tolerance (residual load after first cracking) but the component is arguably not truly fail-safe, because both load paths converge through the same root region. To improve post-damage capability, first spread the tensile load over a longer run-out by increasing the root radius and taper. This is constrained by the loading-hole keep-out on the vertical flange and by the flange width, so very large strut radii are not permissible. Next, add a secondary tie across the root, such as a strap or rib, to create an alternate load path. Finally, raise interlayer tensile and interlaminar shear strength by adjusting print orientation, increasing the number of perimeters, and potentially using higher extrusion temperatures [3, 5].

5.2.2 Automotive Bracket: Observed Failure Modes and Significance

In ??, the load-displacement curve peaks and then drops off sharply, indicating a sudden, critical failure. This critical failure is indicative of a damage-intolerant design, where the system fails catastrophically without prior warning. This is a key aspect of the design of the automotive bracket, which could be improved upon. This failure mode is shown in figure 6. In the figure, we can see a clean horizontal fracture below the joint connecting the strut to the vertical member. This failure is a case of inter-layer delamination. where the applied tensile or shear loads exceeded the adhesion strength between the printed layers. The fracture surface analysis as seen in 6 reveals a clean separation along the horizontal plane. By looking at the part, we can see this is parallel to the print lines along the member. This fracture is characteristic of Mode I, also known as opening mode failure, as the layers of the part have been pulled directly apart. Unlike the progressive failure seen in the aerospace component, which experienced mixed-mode failure, we can see that the automotive bracket's failure was caused by the print orientation. The part was printed so the principal tensile stress was perpendicular to the printed layers, the weakest possible orientation [2]. This explains the sharp, vertical drop in ?? as once the inter-layer delamination began the part had zero residual load-carrying capacity and failed instantaneously.

From a safety standpoint, this is exemplary damage-intolerant behaviour. The design provides no warning of failure and has no fail-safe characteristics. It failed at the predictable location below the joint due to limitations of its manufacturing process. To improve to a more robust design several changes should be made. First, the part's print orientation must be changed to be printed on its side. This aligns the stronger extrusion paths with the direction of the stress and thereby

moving the load away from the weaker inter-layers [2]. Second, to reduce stress concentration, a larger radius fillet should be added to smooth the load transfer between strut and vertical member.

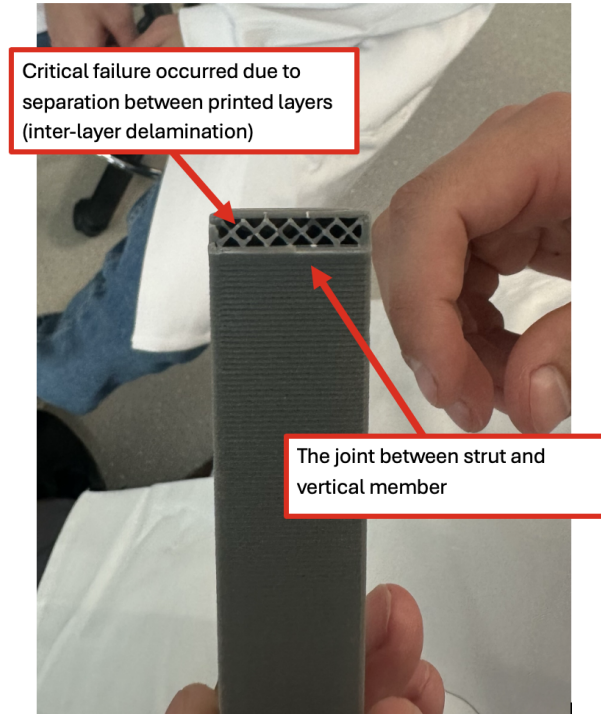


Figure 6: Automotive bracket: observed tension-driven failure at strut root.

5.3 Methods of Further Design Optimisation

Although currently outside the scope of this assignment, further design improvements can be made to both brackets. This section will discuss different techniques that could potentially improve the load-bearing performance of each bracket.

5.3.1 Design for Fused Deposition Modelling

Fused Deposition Modelling (FDM) or Fused Filament Fabrication (FFF) is a common additive manufacturing technology well suited to rapid prototyping, as well as manufacturing complex geometry with significant ease compared to traditional subtractive manufacturing techniques. FDM 3D printing creates parts by extruding a heated thermoplastic in continuous lines across the x-y plane, and building up the geometry of a part layer by layer, laying new beads on top of each previous layer. This process creates highly anisotropic parts, as the thermoplastic has significantly higher strength and stiffness along the length of continuous lines than across layers. Print orientation, amongst a large number of other parameters including; layer height, perimeter count, infill density and pattern, and other printing techniques such as brick layering, can all change the mechanical performance of an FDM 3D printed part significantly.

Both the aerospace and automotive bracket design were observed to fail along layer lines (automotive significantly more than the aerospace) as a result of the principal stress orientation being misaligned with the in-layer higher strength direction of the printed parts. In order to improve performance of each of the brackets, load path analysis should be done to determine the orientation that carries the largest tensile and compressive stresses, and the bracket should be printed in the orientation that aligned the layers with this direction. A limitation of most FDM 3D printers is

that only one orientation can make use of this increased strength, and load paths can vary through parts, meaning a choice must be made for the most critical load path. However, experimentation is currently being done with 3D printers that work in 4 and 5 axis configurations, allowing layer lines to be produced in multiple directions at different areas of a single part.

Additional recommendations for increasing the strength of the bracket through 3D printing techniques include:

- **Increasing perimeter count:** FDM 3D printed parts are printed with multiple lines of extruded filament on the external perimeter of parts, and low density infill inside the perimeters. The perimeters are the primary load bearing part of the structure, and increasing the number of perimeters is directly proportional to increasing the strength of the part.
- **Brick-style print layering:** Brick-style print layering is an experimental technique where instead of having discrete layers across the entire part, layers are extruded at stepped intervals and interwoven - similar to how bricks are laid. This has been shown to increase the strength of parts in the loading direction parallel to layer lines.

As the aerospace and automotive brackets have different requirements and therefore design considerations, additional recommendations are provided specifically for each.

- **Aerospace bracket 3D printing improvements**

- **Different infill patterns:** Changing the infill pattern from *grid* to *gyroid* makes the part have closer to isotropic properties, by printing continuously and weaving infill extrusions in many directions. However, it increases print time, so is less ideal for automotive parts.

- **Automotive bracket 3D printing improvements**

- **Larger nozzle diameter:** By printing with a larger nozzle (compared to the 0.4mm nozzle used in this experiment), larger layer heights can be used, decreasing the number of overall layers and significantly reducing print time.

6 Conclusion

Both brackets met the brief and outperformed the benchmark in strength and stiffness. The aerospace bracket converted the load path to mainly axial through triangulation, which delivered the highest capacity and lowest deflection, but at a longer print time. The automotive bracket used a simpler brace layout that printed faster while still clearing the performance target and beating the benchmark.

Failure consistently initiated at the brace/strut roots with layer separation, indicating stress concentration and highlighting the need for better transitions. Within the limits of FDM (layer adhesion, infill effects, and fixture compliance), the results align with expectations from the design choices.

For applications where strength-per-weight matters, the aerospace bracket is the best option. Where time is the priority, the automotive bracket is the better choice. In future iterations, there would be a larger fillet radius at the roots, and experimentation with print orientation, layer thickness, and perimeter count to improve interlayer strength. These changes target the observed failure modes and should increase the performance of both the aerospace and automotive designs.

References

- [1] 14 cfr 25.303 — factor of safety, 2024. URL <https://www.ecfr.gov/current/title-14/chapter-I/subchapter-C/part-25/subpart-C/section-25.303>. “Unless otherwise specified, a factor of safety of 1.5 must be applied to the prescribed limit load.”.
- [2] Fracture mechanics, 2025. URL <https://mechanicalc.com/reference/fracture-mechanics>.
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Treating the cross section at a general station distance along the horizontal flange, z , as the sum of a rectangle (the flange) and two circles (the cross section of the struts), the y -coordinate of the centroid is given by Equation (8).

$$\bar{y}(z) = \frac{\sum_i y_i A_i}{\sum_i A_i} = \frac{-\frac{1}{2} b_h t_h^2 - 2(t_h + z \tan \theta) \pi R^2}{t_h b_h + 2\pi R^2} \quad (8)$$

Using the centroid position from Equation (8) in the parallel-axis theorem gives the second moment about the moment of inertia about the x -axis is given by Equation (9):

$$I_{xx}(z) = \frac{b_h t_h^3}{12} + b_h t_h \left(-\frac{t_h}{2} - \bar{y} \right)^2 + 2 \left(\frac{\pi R^4}{4} + \pi R^2 (-[t_h + z \tan \theta] - \bar{y})^2 \right) \quad (9)$$

The maximum bending load about the x -axis occurs at the bottom point of the struts, as verified by the Von Mises stress distribution diagrams in Figures ?? under the room temperature conditions, respectively. The maximum bending stress in the L-bracket structure is given by Equation (10):

$$\sigma_{z,\max} = \frac{M_{x,\max}}{I_{xx}} y_{\max} = \frac{P_x z}{I_{xx}} \quad (10)$$

For a unit vertical load ($P = 1$) applied to the horizontal flange at the drill holes, so that $M = z$, the bending stress at the bottom fibre of a strut is given by Equation (11):

$$\sigma_z(z)|_{P=1} = \frac{z}{I_{xx}(z)} \left(-[t_h + R + z \tan \theta] - \bar{y}(z) \right) \quad (11)$$

Given that the struts are tapered with respect to the z -axis, the direct stress along the strut is obtained by projecting σ_z onto the strut axis, via Equation (12):

$$\sigma_{\parallel}(z) = \sigma_z(z) \sqrt{\left(\frac{\partial x}{\partial z} \right)^2 + \left(\frac{\partial y}{\partial z} \right)^2 + \left(\frac{\partial z}{\partial z} \right)^2} \quad (12)$$

$$= \sigma_z(z) \sqrt{1 + \tan^2 \theta + \tan^2 \phi} \quad (13)$$

Yield is assumed when $\sigma_{\parallel} = \sigma_y$, which gives the predicted failure load, evaluated at the governing cross-section $z_{\max}$:

$$P_{\text{fail}} = \frac{\sigma_y}{|\sigma_{\parallel}(z_{\max})|_{P=1}} \quad (14)$$

The averaged section moment of inertia was sampled every $\Delta z = 0.5$ mm across the span (161 stations) as

$$\bar{I}_{xx} = \frac{1}{z_{\max}} \sum_k I_{xx}(z_k) \Delta z, \quad (15)$$

and the tip deflection at the effective horizontal flange length L_{eff} of 87 mm followed from the Euler-Bernoulli beam bending theory:

$$\delta \approx \frac{P L_{\text{eff}}^3}{3E \bar{I}_{xx}}. \quad (16)$$

Figure ?? shows the SolidWorks rendering for bracket 4, designed for automotive use. The subsequent section details the calculation of the maximum failure load and other parameters. Figure ?? shows the engineering drawing for this bracket.

All calculations use units of mm, N, and MPa.

Geometry: All members: 20 mm thick

Horizontal flange: 100 mm long $\times$ 3 mm wide $\times$ 20 mm thick

Vertical flange: 50 mm tall $\times$ 5 mm wide $\times$ 20 mm thick

Diagonal brace: 4.9 mm wide $\times$ 20 mm thick $\times$ 78.9 mm long at 16°

Parameters:

$$l_h = 87 \text{ mm} \quad (\text{horizontal distance to load}) \quad (21)$$

$$L_{\text{eff}} = 100 \text{ mm} \quad (\text{effective length}) \quad (22)$$

$$E = 2100 \text{ MPa} \quad (\text{elastic modulus}) \quad (23)$$

$$\sigma_y = 26.08 \text{ MPa} \quad (\text{yield stress}) \quad (24)$$

Cross-sectional areas for root section analysis:

$$A_{\text{hf}} = 20 \times 3 = 60.00 \text{ mm}^2 \quad (\text{horizontal flange}) \quad (25)$$

$$A_{\text{vf}} = 20 \times 5 = 100.00 \text{ mm}^2 \quad (\text{vertical flange}) \quad (26)$$

$$A_{\text{diag}} = 4.9 \times 20 = 98.00 \text{ mm}^2 \quad (\text{diagonal brace}) \quad (27)$$

$$A_{\text{tot}} = 60.00 + 100.00 + 98.00 = 258.00 \text{ mm}^2 \quad (28)$$

Local centroids with datum at top face (positive y downward). For diagonal brace:

$$\text{Vertical projection} = 78.9 \times \sin(16^\circ) = 21.77 \text{ mm} \quad (29)$$

$$\text{Start position} = 3 + (50 - 21.77) = 31.23 \text{ mm} \quad (30)$$

$$\text{End position} = 3.0 \text{ mm} \quad (31)$$

$$y_{\text{diag}} = \frac{31.23 + 3.0}{2} = 17.12 \text{ mm} \quad (32)$$

Component centroids:

$$y_{\text{hf}} = \frac{3}{2} = 1.50 \text{ mm} \quad (33)$$

$$y_{\text{vf}} = 3 + \frac{50}{2} = 28.00 \text{ mm} \quad (34)$$

$$y_{\text{diag}} = 17.12 \text{ mm} \quad (35)$$

Overall centroid:

$$\bar{y} = \frac{A_{\text{hf}} \cdot y_{\text{hf}} + A_{\text{vf}} \cdot y_{\text{vf}} + A_{\text{diag}} \cdot y_{\text{diag}}}{A_{\text{tot}}} \quad (36)$$

$$\bar{y} = \frac{60.00 \times 1.50 + 100.00 \times 28.00 + 98.00 \times 17.12}{258.00} = 17.70 \text{ mm} \quad (37)$$

Second moment of area using parallel-axis theorem.

Horizontal flange contribution:

$$I_{\text{hf}} = \frac{20 \times 3^3}{12} + A_{\text{hf}}(y_{\text{hf}} - \bar{y})^2 = 15793.2 \text{ mm}^4 \quad (38)$$

Appendix D: Design Optimisation Code

D.1 Aerospace Bracket 1 - Austin Markwick

```
clc;
clear;
close all;

% --- Benchmark Data and Material Properties ---

% Material Properties for PLA
E_GPa = 2.1;           % Elastic modulus (GPa)
sigma_uts_MPa = 36;    % Ultimate tensile strength (MPa)
sigma_y_MPa = 26.08;   % Yield strength (MPa), from report [1]
rho = 1265;            % Density (kg/m^3)

E = E_GPa * 1e9;       % Elastic modulus in Pa
sigma_y = sigma_y_MPa * 1e6; % Yield strength in Pa

% --- 3D Printing Parameters ---
infill_density = 0.4;

% Benchmark Bracket Performance
benchmark.P = 51.27;    % Load at failure (N)
benchmark.m_g = 13.45; % Mass (g)
benchmark.m_kg = benchmark.m_g / 1000; % Mass (kg)
benchmark.delta_failure = 57.76 * 1e-3; % Deflection at failure (m)
benchmark.T = 3600;    % Time for 3D printing (s)

% Cost Function Constants
benchmark.F_aero = 200; % Fuel efficiency constant for aerospace

% Calculated Benchmark Metrics
benchmark.G = benchmark.P / benchmark.m_kg; % Specific strength (N/kg)
benchmark.S = benchmark.P / benchmark.delta_failure; % Rigidity (N/m)
benchmark.V = benchmark.m_kg / rho; % Volume (m^3)

% --- Fixed Design Geometry (in SI units) ---
b_flange = 20e-3; % Horizontal flange width (m)
l_h = 95e-3; % Total horizontal flange length (m)
L_eff = 87e-3; % Effective cantilever length for deflection (m)
z_max = 80e-3; % Max z-distance for strut connection (m)
theta_max_deg = 20; % Max allowable strut angle (degrees)
theta_max = deg2rad(theta_max_deg);

% Volume of the fixed vertical member (constant for all designs)
% Dimensions: 50x20x5 mm with one 3.5mm diameter hole
V_vert_flange = (50e-3 * 20e-3 * 5e-3) - (pi * (3.5e-3 / 2)^2 * 5e-3);

% --- Design Parameter Grids for Optimisation Search ---
% These are the variables we can change to find the best design.
strut_rad = (1.0:0.1:5.0) * 1e-3; % Strut radius [m]
phis = deg2rad(1:0.1:5); % Half-separation angle between struts [rad]
```

```

t_h = (1.0:0.1:10.0) * 1e-3;          % Horizontal flange thickness [m]

% Grid for calculating average moment of inertia along the beam
z_grid = 0:0.5e-3:z_max; % Step every 0.5mm as per report methodology

% --- Optimisation Loop ---
% This section iterates through all possible design combinations to find
    the best one.

fprintf('Starting optimisation search...\n');
best = struct('C1', -inf); % Initialize 'best' structure to store the
    optimal design
counter = 0;
total_iterations = length(strut_radii) * length(phis) * length(t_h);

for R = strut_radii
    for phi = phis
        for th = t_h
            % --- Geometry Calculation for current design ---
            % Theta is determined by the connection points, capped at
                theta_max
            theta = atan((30e-3 - th) / z_max);
            % theta = 20;
            theta = min(theta, theta_max);

            % --- Mass Calculation ---
            V_horiz_flange = b_flange * th * l_h;
            L_strut = z_max * sqrt(1 + tan(theta)^2 + tan(phi)^2);
            V_struts = 2 * pi * R^2 * L_strut;
            V_total = V_vert_flange + V_horiz_flange + V_struts;
            % Adjust mass for infill. Note: This is an approximation as
                the outer
            % shells are typically solid. A slicer gives a more accurate
                mass.
            m_design = 0.01345 + V_struts * rho * infill_density; % Mass
                in kg

            % --- Stress Analysis to find Failure Load (P_fail) ---
            % Calculations are based on the cross-section at z_max where
                moment is highest.
            z = z_max;
            y_bar_zmax = (-0.5*b_flange*th^2 - 2*(th + z*tan(theta))*(pi*R
                ^2)) / (b_flange*th + 2*pi*R^2);
            I_xx_zmax_solid = (b_flange*th^3)/12 + b_flange*th*(-th/2 -
                y_bar_zmax)^2 + ...
                (pi*R^4)/2 + 2*pi*R^2*(-(th + z*tan(theta))
                    - y_bar_zmax)^2;

            % Scale Moment of Inertia by infill density to get effective
                stiffness
            I_xx_zmax = I_xx_zmax_solid * infill_density;

            % Stress is calculated at the bottom-most fiber of a strut
            y_stress = -(th + R + z*tan(theta))-y_bar_zmax;

            % Stress per unit load (P=1N)

```

```

M_per_P = z; % Moment = P*z, so moment per unit load is just z
sigma_z_per_P = (M_per_P / I_xx_zmax) * y_stress;
sigma_parallel_per_P = sigma_z_per_P * sqrt(1 + tan(theta)^2 +
    tan(phi)^2);

% Predicted failure load based on yield strength
P_fail = sigma_y / abs(sigma_parallel_per_P);

% --- Constraint Check ---
% Skip this design if it doesn't meet the minimum load
% requirement
if P_fail < 75
    continue;
end

% --- Cost Function Calculation ---
% Estimate 3D printing time based on mass ratio, which
% accounts for infill density.
T_d = benchmark.T * (m_design / benchmark.m_kg);

% Aerospace cost function C1 (based on specific strength)
C1 = 1.5 * ((P_fail / m_design) / benchmark.G) - (T_d / (
    benchmark.F_aero * benchmark.T));

% --- Update Best Design ---
if C1 > best.C1
    best.C1 = C1;
    best.R = R * 1000; % store in mm
    best.phi = rad2deg(phi); % store in degrees
    best.th = th * 1000; % store in mm
    best.theta = rad2deg(theta); % store in degrees
    best.P_fail = P_fail;
    best.mass = m_design * 1000; % store in grams
    best.T_d = T_d;

    % --- Deflection Calculation (only for the best candidate
    % to save time) ---
    I_xx_samples = zeros(size(z_grid));
    for i = 1:length(z_grid)
        zi = z_grid(i);
        y_bar = (-0.5*b_flange*th^2 - 2*(th + zi*tan(theta))*(
            pi*R^2)) / (b_flange*th + 2*pi*R^2);
        I_xx_samples(i) = (b_flange*th^3)/12 + b_flange*th*(-
            th/2 - y_bar)^2 + ...
            (pi*R^4)/2 + 2*pi*R^2*(-(th + zi*tan
                (theta)) - y_bar)^2;
    end
    I_xx_avg_solid = mean(I_xx_samples);
    I_xx_avg = I_xx_avg_solid * infill_density; % Scale by
    infill
    delta = (P_fail * L_eff^3) / (3 * E * I_xx_avg);
    best.delta = delta * 1000; % store in mm
    best.I_xx_avg_mm4 = I_xx_avg * 1e12; % store in mm^4

    % --- Additional Cost Function & Efficiency Calculations
    ---

```

```

        P_delta_design = P_fail / delta; % Rigidity of current
            design (N/m). Convert delta back to m
        C3 = 1.5 * (P_delta_design / benchmark.S) - (T_d / (
            benchmark.F_aero * benchmark.T));
        epsilon_aero = C1 / 1.5;

        % --- Margin of Safety Calculation ---
        % MOS = (Allowable Stress / Actual Stress) - 1
        % Actual Stress = Calculated stress at 75N load * Factor
            of Safety (1.5)
        stress_at_75N = abs(sigma_parallel_per_P) * 75; % Stress
            in Pa
        MOS = (sigma_y / (stress_at_75N * 1.5)) - 1;

        best.C3 = C3;
        best.epsilon_aero = epsilon_aero;
        best.MOS = MOS;
    end

    counter = counter + 1;
    if mod(counter, 500) == 0
        fprintf('...processed %d / %d iterations.\n', counter,
            total_iterations);
    end
end
end
end

% --- Display Results ---
fprintf('\n--- Optimisation Complete ---\n');
if isinf(best.C1)
    fprintf('No design found that meets the minimum load requirement of 75
        N.\n');
    fprintf('Consider expanding the parameter grids or checking
        constraints.\n');
else
    fprintf('Optimal Design Parameters Found (at %.0f%% infill):\n',
        infill_density*100);
    fprintf(' - Strut Radius (R):           %.2f mm\n', best.R);
    fprintf(' - Flange Thickness (t_h):       %.2f mm\n', best.th);
    fprintf(' - Strut Half-Angle (phi):             %.2f deg\n', best.phi);
    fprintf(' - Strut Angle (theta):                 %.2f deg\n', best.theta);
    fprintf('\nPredicted Performance:\n');
    fprintf(' - Strength Cost (C1):                   %.4f\n', best.C1);
    fprintf(' - Rigidity Cost (C3):                   %.4f\n', best.C3);
    fprintf(' - Aero Efficiency (eps_a):              %.4f\n', best.epsilon_aero);
    fprintf(' - Failure Load (P_fail):                %.2f N\n', best.P_fail);
    fprintf(' - Margin of Safety (at 75N):            %.2f\n', best.MOS);
    fprintf(' - Total Mass:                           %.2f g\n', best.mass);
    fprintf(' - Deflection at Failure:                 %.2f mm\n', best.delta);
    fprintf(' - Avg. Moment of Inertia:               %.2e mm^4\n', best.I_xx_avg_mm4
    );
    fprintf(' - Est. Printing Time:                   %.0f s\n', best.T_d);
end

```
